

# KISHAN DEKA

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## EDUCATION

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| National Centre for Nuclear Research, Warsaw<br><i>PhD Student (Supervisor : dr. hab. Paweł Bielewicz)</i>                           | 2022 – Present               |
| National Institute of Science Education and Research, Bhubaneswar<br><i>Integrated B.Sc.-M.Sc. in Physics (Mathematics as minor)</i> | 2017 – 2022<br>CGPA: 8.89/10 |

## RESEARCH INTERESTS

Weak lensing reconstruction of the cosmic microwave background (CMB)  
Optimal de-lensing methods with external large-scale structure tracers  
Systematic biases in de-lensing and in primordial B-mode amplitude constraints  
Galactic foreground emission models and component separation methods  
Impact of systematics in cross-correlation studies between CMB and galaxy surveys

## PAST RESEARCH EXPERIENCES

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| Developing PSF photometry pipeline for the WALOP instrument.<br><i>Guides : Dr. Tuhin Ghosh and Prof. A. N. Ramaprakash (Master's Thesis)</i> | Aug 2021 – June 2022<br>NISER Bhubaneswar |
| Signature of magnetic helicity in thermal dust polarization<br><i>Guide : Dr Tuhin Ghosh</i>                                                  | July 2020 – May 2021<br>NISER Bhubaneswar |
| Viscous Landau hydrodynamics in heavy-ion collisions<br><i>Guide : Dr. Amaresh Jaiswal (Report)</i>                                           | May 2019 – Oct 2019<br>NISER Bhubaneswar  |

## SELECTED PUBLICATIONS

ORCID : [0000-0002-8229-4326](https://orcid.org/0000-0002-8229-4326)

Deka, K. & Bielewicz, P., *Galactic foreground residual biases in CMB lensing convergence reconstruction and delensing of B-mode maps*, accepted in A&A (Feb 2026), DOI : [10.1051/0004-6361/202557977](https://doi.org/10.1051/0004-6361/202557977)

Adhikary, J., Batra, A., Deka, K., Joaquim, F.R., *Sterile neutrino Dark Matter in the minimal Dirac Seesaw*, arXiv preprint (March 2026), DOI : [10.48550/arXiv.2603.20145](https://doi.org/10.48550/arXiv.2603.20145)

Biswas, D., Deka, K., Jaiswal, A., Roy, S., *Viscosity, nonconformal equation of state, and sound velocity in Landau hydrodynamics*, Phys. Rev. C **102**,014912 (2020). DOI: [10.1103/PhysRevC.102.014912](https://doi.org/10.1103/PhysRevC.102.014912)

## SEMINARS/TALKS

- *Challenges for CMB B-Modes: Lensing Reconstruction, Subtraction, and Diffuse Foregrounds*, at SISSA, Trieste, Italy, 2026 (invited)
- *Biases in CMB lensing convergence reconstruction and optimal delensing of B-modes with next-generation surveys*, at LAM Marsellei, France, 2025 (guest seminar)
- *Optimal delensing of CMB B modes with stage-IV CMB and galaxy survey*, at XI Fundamental Cosmology in Santander, 2025
- *Delensing with tomographic cross-correlation between LSST and CMB-S4 survey*, at LSST@Europe7, 2025
- *Delensing of CMB B-modes with tomographic cross-correlation between LSST and CMB-S4 survey*, at Dark-Map in Budapest, 2025

- *Impact of galactic foregrounds in delensing of CMB B-mode polarisation*, CAMK PAN, Warsaw, 2025 (invited)
- *Impact of galactic foregrounds on delensing of CMB maps*, at Cosmoverse@Istanbul, 2025
- *Galactic foreground bias in CMB lensing reconstruction*, SPS Seminar, NISER Bhubaneswar, India, 2024 (alumni talk)
- *Galactic foreground bias in CMB lensing convergence reconstruction*, at Cosmology in Miramare, 2023
- *PSF modeling for the WALOP instrument*, Astrophysics(BP4) seminar, NCBJ Warsaw

## CONFERENCES/SCHOOLS

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- Rencontres de Moriond in Cosmology, March 2026 (poster)
- XI Meeting on Fundamental Cosmology, Santander, Spain, Nov. 2025 (talk)
- LSST@Europe, Poznan, Poland, Sept. 2025 (talk)
- Advanced School on Precision Cosmology and Dark Matter, GGI, Florence, Italy, Aug. 2025 (school)
- DarkMap - Mapping the Dark Universe, Budapest, Hungary, Sept. 2025 (talk)
- Cosmoverse@Istanbul, Turkey, June 2025 (talk)
- PAiP-2025-Particle Physics in Poland, Warsaw, Poland, Feb. 2025 (flash talk)
- Cosmoverse@Krakow, Poland, July 2024 (poster)
- Cosmology in Miramare, at SISSA Trieste, Italy, Aug. 2023 (talk)
- EAS Annual meeting 2023, ICE Krakow, Poland, July 2023 (poster)
- 9<sup>th</sup> IIST Astronomy and Astrophysics School, at IIST Thiruvananthapuram, Dec. 2020 (school)
- Dynamics of QCD Matter, at NISER Bhubaneswar, Aug. 2019
- Vijayoshi Science Camp, at IISER Kolkata, Dec, 2017 (winter school)

## COLLABORATIONS/VISITS

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- Member of DESC, LSST collaboration (PI : Paweł Bielewicz)
- Member of CMB-S4 collaboration (PI : Jacques Delabrouille)
- Visiting researcher in LBNL, Berkeley, CA (Sept – Nov, 2024)
- Visiting researcher in SISSA, Trieste, Italy (March, 2026)
- Collaborative visit in LAM, Marseille (Nov, 2025)

## FELLOWSHIPS/AWARDS

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| • <b>STSM Grant, Cosmoverse (CA21136)</b> , COST Action, EU            | <b>March 2026</b>  |
| • <b>STER Fellowship (BPI/STE/2021/1/00033/U/00001)</b> , NAWA, Poland | <b>2023 – 2024</b> |
| • <b>INSPIRE Fellowship</b> by Dept. of Science and Technology, India  | <b>2017 – 2022</b> |

## TEACHING EXPERIENCE

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- Teaching assistant in **Classical Electrodynamics** with Prof. Andrzej Hryczuk, NCBJ, Warsaw, Fall 2025 (post-graduate level)

## COMPUTATIONAL SKILLS

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**Languages:** Python (proficient), Fortran90, C++

**Packages:** healpix, astropy, camb, pysm3, emcee, plancklens, lenzpy, namaster, glass, cobaya

**Others:** Mathematica, Latex, Git, Linux

## REFERENCES

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Dr. hab. Paweł Bielewicz, NCBJ, Warsaw, Poland

Dr. Tuhin Ghosh, NISER, Bhubaneswar, India

Prof. Carlo Baccigalupi, SISSA, Trieste, Italy